



Understanding the science-policy interface: Case studies on the role of information in fisheries management



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ABSTRACT

Case studies of the Canada Department of Fisheries and Oceans (DFO), Northwest Atlantic Fisheries Organization (NAFO), and the Food and Agriculture Organization of the United Nations (FAO) highlighted factors influencing scientific communication that are contingent on the characteristics of the many dynamic and iterative science-policy interfaces among decision-makers, scientists, and other stakeholders. Direct observations of 15 scientific and management meetings coupled with interviews with 78 scientists and managers revealed aspects of the information pathways, i.e., production, communication, and use of scientific information in these organizations. Unique features of decision-making and information use enable the production of credible, relevant, and legitimate information in each organization, including trade-offs in these attributes to support fisheries governance objectives. For instance, defined processes for producing scientific advice embedded in fisheries management authorities, such as DFO and NAFO, ensure uptake of information in decision-making. As a boundary organization, FAO bridges science and policy-making groups among its member countries. The demand for scientific advice, policy development, and trade aspects are primary drivers in the information pathways. However, organizational aspects such as dispersed units and inadequate communication persist as barriers to information flow. Across the geographic scales of the three organizations, stakeholders apart from government scientists and policy-makers, e.g., the fishing industry, non-governmental organizations (NGOs), and the public, are increasingly involved in the information pathways. Insights about the information pathways can equip the organizations to evaluate or modify practices to increase the uptake of information in decision-making as fisheries management issues and considerations become more complex.

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1. Introduction

Scientific information in some jurisdictions with well-assessed fisheries has been assimilated into fisheries management decisions to address the declines in fish populations and catches, biodiversity, and associated livelihoods. However, there are numerous instances where use of available information in fisheries management and policy-making contexts to mitigate problems is not obvious (Costello et al., 2012; FAO, 2014e; Holmes and Lock, 2010; Worm et al., 2009). Questions about information use (or non-use) have been considered from two perspectives. First, limited use of scientific information by governments in policy- and decision-making is attributed to impediments with information flow at the science-policy interface (Dicks et al., 2014; Lalor and Hickey, 2013;

Soomai et al., 2013). Second, the role that scientific information plays in policy-making and decision-making processes is rarely described; information is commonly viewed as implicit in these processes and it is often not recognised as playing a critical and unequivocal function in its own right (MacDonald et al., 2016).

The limited visible profile of the functions of available scientific information within policy and decision-making communities and the lack of understanding of such information use is not unique to fisheries management. Increasingly, questions about information use are being raised and examined in the context of evidence-based policy-making where decisions are expected to be made based on the available scientific information (Gluckman, 2013; MacDonald et al., 2015; Nurse-Bray et al., 2014; Nutley et al., 2007). At the same time, international commitment to strengthen communication at the science-policy interface and to enhance evidence-based decision-making at all jurisdictional levels has also intensified (UNGA, 2012; UN DESA, 2013).

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This paper empirically examines the role(s) of scientific information in the science-policy interface in decision-making for fisheries management in three governmental organizations: the Canada Department of Fisheries and Oceans (DFO), the Northwest Atlantic Fisheries Organization (NAFO), and the Food and Agriculture Organization of the United Nations (FAO) (Soomai, 2015). This research reveals the multidimensional processes at the science-policy interface by which scientific information is produced, communicated, and considered in policy decisions, and the enablers and barriers to this activity. Insights from the existing literature on decision-making and information use set the context for the study. The findings can equip fisheries governmental organizations, among other organizations, to evaluate or modify practices to increase the uptake of information in decision-making. The new knowledge can enable interested stakeholders to determine the most appropriate entry point to introduce evidence and to participate in decision-making processes.

2. Insights from previous studies on the science-policy interface and information use

The science-policy interface is a complex phenomenon but in its simplest form it can be characterised by communication between information production (science) and information use (policy) (Ascher et al., 2010; Liverani et al., 2013; Mitchell, 2010; van den Hove, 2007). The interface encompasses many social processes related to policy- and decision-making. Understanding the role of scientific information is particularly challenging given the complexities of modern societal and environmental issues. Governments' use of information is often grounded in a range of internal and external factors related to institutional and organizational aspects, the characteristics of the actors involved in policy processes, or characteristics of the information itself (Head, 2016; Likens, 2010; Mitchell et al., 2006; Mol, 2008). Uptake of fisheries scientific information into management advice by governmental organizations may be influenced by governance models, political regimes, the geographic region, information management cultures of science and management domains, and personal and institutional interests and values of multiple stakeholders (Cochrane, 2002; Cossarini et al., 2014; Delaney and Hastie, 2007; Hastie, 2008; Soomai et al., 2011; Wilson, 2009). The level of technical details in scientific information provided as advice and the degree of scientific uncertainty can influence government decision-making and public engagement in policy-making (Dankel et al., 2012; Hauge, 2011; Rosenberg, 2007; Wardekker et al., 2008). Decision-making extends beyond the interaction of government scientists and policy-makers and involves other stakeholders including resource users, NGOs, and the public; these groups often exist as networks (Clark et al., 2011; Hartley and Glass, 2010; Irvine, 2009; Soomai et al., 2013). Additionally, scientific knowledge interacts with other types of knowledge, e.g., local knowledge, and may compete with other kinds of information, including that from economic and social sciences, in decision processes (Cicin-Sain and Knecht 1998; McNie, 2007). With a global trend towards ecosystem-based management to balance diverse societal objectives with fisheries management, fisheries publications by governmental and non-governmental organizations (NGOs) have become even more prolific and more interdisciplinary (Garcia et al., 2003; Levin et al., 2009; UN, 1972, 1992). Political agendas and attempts to maintain the neutrality of science can also affect information flow at the interface (Guston, 2001; Koetz et al., 2011; Pielke, 2007; Sarewitz, 2014). Such diverse factors can create opportunities or challenges for the communication of scientific information to policy-makers and its use upon receipt.

Many models and perspectives on decision-making and information use assume evidence-based policy-making and

describe a linear or science-based interface and a collaborative or participatory approach (Bremer and Glavovic, 2013; Bulkeley, 2005; Knol, 2010; Sarewitz and Pielke, 2007; Stojanovic et al., 2009; Young et al., 2013). The pathway(s) of information in policy-making can follow an "ideal" or linear process as researchers ask the relevant questions, plan and conduct research, and then disseminate the scientific advice directly to the policy-maker (Glasziou and Haynes, 2005; Knott and Wildavsky, 1980; Landry et al., 2001; Weiss, 1979). Alternatively, in an "enlightenment model," the links between research and policy are less direct as information accumulates over time and permeates gradually into the policy process (Weiss, 1977). Nutley et al. (2007) describe an iterative process where information flows in a continuum between direct (instrumental) use of scientific information for immediate problem-solving and indirect (conceptual) use where information serves an "enlightenment" purpose over time. Three attributes of information, i.e., credibility, relevance (or salience), and legitimacy, typify information use in the science-policy interface models (Cash et al., 2003; Koetz et al., 2011; Mitchell et al., 2006; Sarkki et al., 2014, 2015).

In-depth studies, as described in this paper, provide empirical evidence of the characteristics of the science-policy interface, particularly with regard to the range of factors and models that have been defined above. Studies on the role of information are becoming more important to organizations where funding for research is quite limited and accountability for information production is needed (EIUI, 2016).

3. Methods

Data were gathered in three-month periods during which interviews were conducted and meetings were observed in each of DFO, NAFO, and FAO between September 2013 and July 2014. DFO is the lead federal agency responsible for developing and implementing national fisheries policies and programs in Canada (DFO, 2016b). NAFO is an intergovernmental fisheries science and management body for the Northwest Atlantic region (NAFO, 2016). FAO, a specialized agency of the United Nations, is the foremost global fisheries management organization providing neutral advice to its members (FAO, 2016a). These governmental organizations operate at different jurisdictional levels but their activities are also inter-related. Canada as a Contracting Party to NAFO is represented by DFO. Similarly, DFO is the official contact organization for Canada as a member country of FAO. The three governmental organizations are prolific producers of fisheries scientific information that is published primarily as grey literature, i.e., published by the organizations themselves and not by a commercial publisher (GreyNet, 2016).

Four specific questions were addressed in each organization: (1) What are the drivers in producing, communicating, and using marine fisheries information by the organizations? (2) What are the information management strategies of the organizations, particularly with regard to communication? (3) What are the institutional and social enablers and barriers in the organizations to scientific information use at the science-policy interface? and (4) Who are the actors in the information pathways and what are the information behaviours (e.g., information seeking and sharing) of the various actors in each organization?

Seventy-eight interviews of scientists and decision-makers in the three organizations (Table 1) were conducted using a semi-structured questionnaire that consisted of open-ended questions to obtain information in the four areas listed above. The research within DFO focused on the DFO-Maritimes region (DFO-MR), the administrative region of DFO on the Atlantic coast. Research within NAFO focused on the participation of Canada as a Contracting Party to NAFO. The research primarily involved government scientists

Table 1

Interviews conducted in the Canada Department of Fisheries and Oceans (DFO), the Northwest Atlantic Fisheries Organization (NAFO), and the Food and Agriculture Organization of the UN (FAO). (N = Number of respondents).

Case Studies	Interview Participants	Scientists	Managers
DFO (N = 26)	Maritimes Region (DFO-MR) Regional Director General's Office; Science, Fisheries Management, and Ecosystem Management sectors National Capital Region (DFO-NCR) Office of the Assistant Deputy Minister in two sectors – Ecosystems and Oceans Science, Ecosystems and Fisheries Management	6 1	18 1
NAFO (N = 19)	DFO-Newfoundland and Labrador Region (DFO-NLR) Science and Fisheries Management sectors DFO-Maritimes Region (DFO-MR) Science sector – Marine Ecosystems and the Oceanography and Climate Change branches DFO-National Capital Region (DFO-NCR) Office of the Assistant Deputy Minister, Ecosystems and Oceans Science Sector; Office of the Senior Assistant Deputy Minister, Strategic Policy NAFO Secretariat NAFO Observers Non-governmental organizations	7 2 1 1 2	2 2 2
FAO (N = 33)	Fisheries and Aquaculture Department Six branches – Statistics and Information; Marine and Inland Fisheries; Policy, Economics, and Institutions; Products, Trade and Marketing; Fishing Operations and Technology; and Aquaculture	33	

and managers, including senior managers or science advisors who interacted with the policy-making communities. Persons involved in the production of science advice and management decisions were invited by email; they were identified from the organizational staff mailing lists and from content analysis of reports of meeting. The interviews ranged from 20 to 90 min, averaging 45 min, and were either audio recorded or notes were taken and responses were later transcribed.

Direct observations were conducted at 15 science and management meetings (Table 2) of senior decision-makers (policy-makers), fisheries managers, and scientists of the organizations and other planned events where there was interaction between decision-makers and the scientific community. These meetings had formal agendas and ranged from two hours or one day, to others scheduled for one to two weeks. At such meetings, the interaction among the decision-makers and scientists was observed with regard to communication of the information sources used by each group.

The research used a mixed method approach, largely qualitative methods. Mixed research methods provided a basis for verifying data, and supplemented data from the different methods. Direct observations complemented the data collected in the interviews of scientists and managers and also corroborated some of the responses obtained in the interviews. Key publications by scientists and decision-makers were read to provide background on organizational activities and the actors involved.

Alpha-numeric identifiers were assigned to the interview responses to ensure the anonymity of the interviewees. The qualitative data from interviews and direct observations were manually coded, i.e., themes and concepts were identified to describe the information pathways (production, communication, and use of information) using methods for content analysis (Coffey and Atkinson, 1996; Kondracki et al., 2002). Drivers, enablers, and barriers in the information pathways were revealed, as well as the characteristics of the science-policy interface that

Table 2

Direct observations conducted in the Canada Department of Fisheries and Oceans (DFO), the Northwest Atlantic Fisheries Organization (NAFO), and the Food and Agriculture Organization of the UN (FAO). (N = Number of meetings).

Case Studies	Meetings at which Direct Observations were Conducted
DFO (N = 8)	Canadian Scientific Advisory Secretariat (CSAS) Regional Peer Review meetings (snow crab, scallop, ecologically and biologically significant areas (EBSAs) (DFO, 2013, 2014a, 2014b) Fisheries advisory committee annual meetings (hagfish, shrimp) Inter-departmental meetings (Science, Management, and Policy and Economics sectors)
NAFO (N = 3)	35th NAFO annual meeting (NAFO, 2013a) 6th meeting of the Working Group on Ecosystem Science and Assessment (WGESA) (NAFO, 2013b) 1st NAFO Scientific Council-Fisheries Commission Working Group on Risk-based Management Strategies (WGRBMS) (NAFO, 2014)
FAO (N = 4)	Development of a global vulnerable marine ecosystems (VME) database (FAO, 2014b) Workshop on "Sustainable Fisheries Management and Biodiversity Conservation of Deep-sea Living Marine Resources and Ecosystems in the Areas Beyond National Jurisdiction (ABNJ Deep Seas Project)" (FAO, 2014d) 31st session of the Committee on Fisheries (COFI) (FAO, 2014c) Internal meeting of FAO staff related to the implementation of the "FAO Voluntary Guidelines for Securing Sustainable Small-Scale Fisheries in the Context of Food Security and Poverty Eradication" (FAO, 2014f)

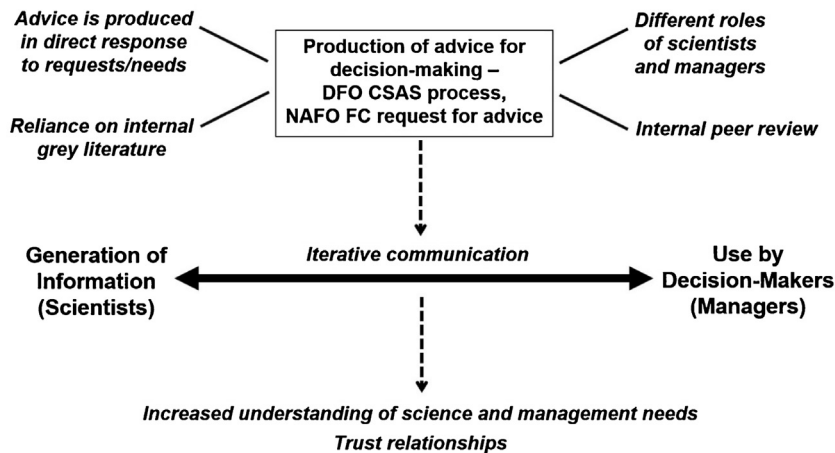


Fig. 1. Conceptual diagram of the well-defined processes for production, communication, and use of scientific information in DFO and NAFO.

create credible, relevant, and legitimate information for decision-making.

4. Results

4.1. Information management strategies of the organizations

4.1.1. Defined processes in DFO and NAFO

The science-policy interface in operational fisheries management in DFO, observed in the Canadian Scientific Advisory Secretariat (CSAS) process, and in NAFO, with regard to the NAFO Fisheries Commission (FC) annual request for scientific advice, are defined processes for the production of scientific and management advice. The DFO CSAS process and the NAFO FC request for advice follow sequences of events in national and regional fisheries management respectively (DFO, 2016a; NAFO, 2014). Important features of these processes include: the separation of science and management, characterised by different roles of scientists and managers; the advice (supply) is provided in response to requests (demand); there is continuing iterative communication, i.e., frequent and ongoing, formal and informal dialogue between scientists and managers, that facilitates increased understanding of science and management needs; and the management advice relies primarily on grey literature produced by the organizations (Fig. 1). Respondents emphasized the firm directives for scientists to provide scientific advice only; developing recommendations was the role of the manager. Notwithstanding this separation, iterative communication between managers and scientists, in the production of scientific advice, maintains scientific integrity that is essential for policy-making. Communication helps to establish trust between scientific and decision-making communities. Scientific information is used in direct problem-solving as the collection of new scientific knowledge feeds into decision-making and the organizational and governance structures are designed for this science-based approach to decision-making.

The authority of the organizations and the embeddedness of the information production and the management processes within these organizations ensure credibility, relevance, and legitimacy of the information for operational decision-making. The processes include protocols that are developed in collaboration with scientists and decision-makers. Fisheries decision-makers generally do not question the relevance or credibility of the advice coming from assessments produced within their organizations as the advice is tailored to inform their specific

management objectives for the fish stocks and ecosystems in the Northwest Atlantic. The transparency of the decision-making process was highlighted by the public availability of the scientific advice. For instance, online copies of NAFO's adopted papers and reports demonstrate transparency of fishing operations. Likewise, the DFO CSAS technical and meeting report series is an online record of the requests for advice and its subsequent production.

4.1.2. FAO acts as a boundary organization

FAO acts as a boundary organization to bridge science and policy-making communities in its member states. Nine characteristics of FAO as a boundary organization relate to organizational structure and information management and are depicted in Fig. 2. Recognised as a global authority, the FAO contributes to resolving complex global fisheries management issues by setting the agenda of the international community, convening related meetings and negotiations, developing normative codes of conduct or soft law, and influencing state politics on issues. Political considerations inevitably occur, given that decision-making is completed by senior-government officials and politicians. However, the organization operates under the principle of political neutrality in its interactions with its member countries. The FAO, along with other UN agencies, form a global network for integrating national and regional advice on the biological,

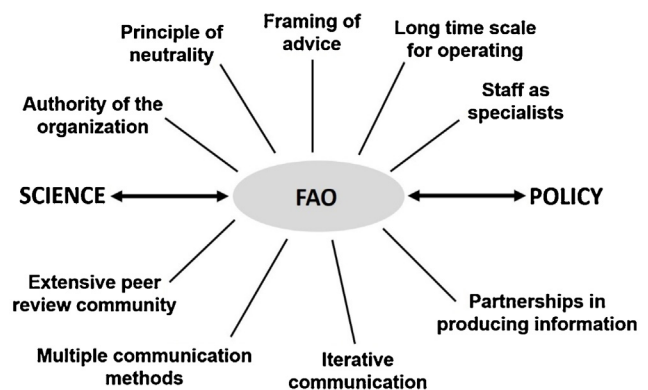


Fig. 2. The characteristics of a boundary organization, illustrated by FAO. (General features of the organization are presented in the top half of the figure and information and communication features in the bottom half).

ecological, economic, and social aspects of fisheries management issues into policy decisions. Fisheries management issues framed in a social and economic context, rather than biological, can influence the attention that they require. Issues cast in a sustainable economy or a human rights context often enhance the relevance of information. For instance, the “Blue Growth Initiative” is important for mobilizing resources and advocacy related to the “green economy,” i.e., sustainable use and conservation of resources in an economic, social, and environmental context (FAO, 2014a; HLPE, 2014).

FAO employs specialists – described as neither scientists nor policy-makers – who “translate” technical information for policy-making communities and other non-technical groups, such as the fisheries community and civil society, to engage them in policy networks. These specialists play a bridging role between national scientists and managers in different fora, e.g., at the UN General Assembly meetings and FAO’s Committee on Fisheries. With regard to its key information products such as the Fisheries and Resources Monitoring System (FIRMS), a global inventory of marine resources and fisheries, a specialist noted that “we spend a lot of time trying to explain the need to scientists, fisheries ministers, and a wide range of people, the importance of having a very easy and effective way of delivering information” (FAO 27). Information production in FAO generally occurs on an ad hoc and project-driven basis. As such, information production has flexible cycles that are contingent on the level of global awareness, concern, and action to resolve an issue, including the availability of funds to coordinate the process. FAO’s bridging role between science and policy groups is vital to promoting the uptake of its information at national levels.

Interaction with diverse peer review communities, including scientists, policy-makers, the fishing industry, NGOs, and civil society groups, is a common approach to producing information in FAO. Partnerships with these stakeholders operating at national, regional, and international levels are increasingly becoming a major component of the organization’s work program, for example, the Areas Beyond National Jurisdiction (ABNJ) and the iMarine projects (FAO, 2014d, 2016c). Comprehensive datasets and publications are produced by such partnership projects where multiple groups can provide feedback, thereby addressing

scientific uncertainty. Multiple communication tools are utilised to share the different information products with the wide range of partners. Formal and informal networks formed in these partnerships build trust among scientists and decision-makers within member countries.

4.2. Characteristics of the science-policy interface in the organizations

The characteristics of DFO, NAFO, and FAO in the context of existing models of decision-making and information use at the science-policy interface are summarized in Table 3.

4.2.1. Models of decision-making and information use at the science-policy interface

The defined structured processes for operational decision-making follow a linear decision model (Table 3). The information pathways are also characteristic of earlier models where information flow occurs as a linear series of events when the request for advice by managers prompts the production of advice by scientists which is subsequently forwarded to managers leading to decision-making. Information use in DFO and NAFO was direct or instrumental as scientific advice was used in operational decision-making for fisheries management, e.g., for setting license conditions, total allowable catches, and guiding fisheries negotiations. Decision-making in the DFO CSAS and the NAFO FC processes primarily uses fisheries science and no other types of information, such as local knowledge.

The science-policy interface in FAO is best defined by a participatory or collaborative interface model (Table 3). The organization commonly calls itself a “knowledge organization,” as evident in its enormous efforts to integrate global datasets from a range of disciplines, such as biology, economics, and policy. Information use in FAO is generally “knowledge-driven,” where a product serves an educational purpose for the intended audiences in the member countries. The information pathway in FAO follows a continuum model where information use cycles between direct and indirect use depending on the type of information product and its intended use, purpose, and audience. Information from the global statistical databases is used directly, often quickly, by the fishing industry to obtain and maintain environmental

Table 3
Characteristics of the science-policy interface in DFO, NAFO, and FAO.

Characteristics	DFO	NAFO	FAO
Decision-making	Linear model – DFO CSAS and the NAFO FC annual request for fisheries management advice; Emphasis on science (Pielke, 2007); Two communities – scientists and managers (Koetz et al., 2011)		Collaborative (participatory) model – Multidisciplinary information; Extended community, i.e., multiple stakeholder groups (Bremer and Glavovic, 2013)
Information Use (Nutley et al., 2007; Weiss, 1979)	Linear model – DFO CSAS and NAFO FC annual request for advice for making management decisions		Continuum model – Cycles between policy development and information production
Category of Use (Nutley et al., 2007)	Direct (instrumental) use – DFO-MR national operational decision-making for annual fisheries management	Direct use – Negotiations and management by Contracting Parties in the Convention Area	Direct use – Global and national policy development; Indirect (conceptual) use – Awareness building and change in attitudes of decision-makers
Typology of Information Use (Weiss, 1979)	Scientific information is used for “problem solving” in fisheries management “Interactive” typology – Communication between decision-makers and information producers is an iterative process, i.e., ongoing and frequent		Production is “knowledge-driven” with an “enlightenment purpose”
Trade-offs in applying the attributes of information	“Matching up” – Advice is provided in response to management questions but other information sources and issues can be overlooked		“Production-communication” – Production to maintain the demand and supply with limited processes for communication
	“Consultation-peer review” – Wide peer community increases dialogue and legitimacy of the information but prevents rigorous review	No “consultation-peer review” trade off was noted	“Consultation-peer review” – Wide peer community increases dialogue and legitimacy of the information

certification. Conversely, the uptake of FAO's information into national decision-making may emerge after many years due to increased awareness of issues, as seen in FAO's ongoing support to its member countries in adopting the 1995 *Code of Conduct for Responsible Fisheries* as a basis for the ecosystem approach to fisheries management (EAF) (Coll et al., 2012). The information contained in related publications can result in increased understanding, a change in attitudes of policy-makers and practitioners in the member countries, and a change in political processes; the information thereby permeating over time into national decision-making.

4.2.2. Trade-offs in applying the attributes of information

In the “matching-up” trade-off in DFO and NAFO, credibility and relevance of information is favoured over legitimacy to ensure that the appropriate scientific information is available for annual operational decision-making for fisheries management. The credibility and relevance of the information produced in the DFO CSAS and NAFO FC processes are assured when specific questions are posed by the managers to the scientists, the advice, i.e., the answers, is provided by the scientists, and the management decisions are made by managers based on the scientific advice (i.e., matching up).

In the “consultation-peer review” trade-off, peer review of scientific publications prepared by DFO scientists is carried out in a “consultation” type meeting involving multiple stakeholders, e.g., the fishing industry, NGOs, academic institutions, government scientists, and managers. Broadening the scope of the peer review community addresses concern about scientific uncertainty about the scientific data by the different stakeholders and facilitates increased understanding of the risk involved in decision-making, thereby increasing the legitimacy of the information that is tabled. A similar trade-off was not observed in NAFO where the peer review of the scientific advice involved only the fisheries and ecosystem scientists of the Contracting Parties who were participating in the Scientific Council. Other groups are not invited to participate, likely because of the underlying confidentiality of the negotiating positions of the Contracting Parties.

The “production-communication” trade-off observed in FAO facilitates its role as a boundary organization in knowledge translation. FAO assigns priority to production of information, however, the organization is increasingly relying on partnerships with external bodies to communicate its messages and to promote awareness of contemporary issues in policy- and decision-making communities. This trade-off is largely due to austerity measures, staff workloads, and the lack of communication staff. For instance, in collaboration with environmental NGOs, the organization vigorously promotes the implementation of the EAF, the management of deep sea fisheries, and the elaboration of provisions of the small-scale fisheries guidelines in national policy within its member countries (FAO, 1993, 1995, 2014f). The information products may not be relevant to the specific needs of all member countries; the “production-communication” trade-off favours credibility and legitimacy over relevance.

5. Discussion

5.1. Drivers of the production of information

The demand for scientific information for decision-making is clearly a main driver in information production in the DFO CSAS process and the NAFO FC annual request for advice. Likewise, FAO plays a critically important role in setting international agendas that drive the production of information. Structured processes embedded within governmental organizations, such as the DFO CSAS and the NAFO FC annual request for advice, provide credible,

relevant, and legitimate information in a transparent manner for operational decision-making. Information produced by organizations recognised as authorities due to their mandates and their internal experts is considered to be credible. DFO's and NAFO's own fisheries scientific publications are preferred in operational decision-making – rather than information in peer-reviewed scholarly journals – because they are timely, i.e., the reports are produced in an annual cycle, and are relevant because they are produced in direct response to the fisheries management needs and questions. Grey literature is preferred – over individually authored publications – as background information for negotiations and at high level policy-making fora in the UN system, e.g., the General Assembly. Such publications are considered the most credible information for use in international governance settings as they are believed to be unbiased and not associated with an individual, not linked to a member country, and not affiliated with other organizations.

Policy development at the different jurisdictional levels is also an important driver in the information pathways. As global and contemporary fisheries management issues continue to become more complex, intergovernmental and national governmental organizations are striving to stay abreast of global policy developed within FAO and the wider UN system. For instance, NAFO's work on vulnerable marine ecosystems (VMEs) is strongly influenced by the UN General Assembly Resolution UNGA 61/105 that initiated a chain of activities addressing human impacts on ocean bottom habitats (FAO, 2009). Such UNGA resolutions can also be viewed as a culmination of the complex interplay of relevant information produced at the international (e.g., FAO), regional (e.g., NAFO), and national (e.g., DFO) levels. FAO's information is intended to increase international awareness of fisheries issues, to call for fisheries management action in its member countries, and to guide international, regional, and national policy-making. The FAO's *Code of Conduct for Responsible Fisheries* clearly demonstrates this role as the Code is widely recognised as the global standard for fisheries management, serving as a basis for reviewing and adopting national fisheries legislation that promotes ecosystem approaches (FAO, 1995).

Government scientists and decision-makers (managers) are the key participants in information production in the governmental organizations. However, NGOs, the fishing industry and civil society, were increasingly engaged in policy- and decision-making processes. International and local NGOs participating as accredited observers in NAFO and as invited “experts” at DFO CSAS review meetings were generally research-oriented and assisted in the production of advice for policy-makers. NGOs based in Nova Scotia were a conduit for information on global agendas, reaching the staff of administrative regions, such as the DFO-MR, who do not generally attend international meetings. NGOs are key partners in information dissemination to FAO member countries, in driving activities under global projects, and in strengthening fisher networks, consequently increasing the lobbying capacity of the industry and civil society, e.g., the adoption of the FAO small-scale fisheries guidelines (FAO, 2014f). NGOs can also lobby national governments and can criticize national governments' actions or inaction – something that FAO cannot do openly as a UN agency. Advances in the organization of the fishing industry and the educational level of its members facilitate participation in the technical DFO CSAS science peer review meetings. Governmental partnerships with the fishing industry and groups representing civil society are an increasingly important means of information production due to limited fiscal resources for research and publications. From an economic perspective, the fishing industry demands information on species sustainability, such as stock status and global capture fisheries production statistics, in order to maintain markets. Regulation of

international trade in fish and fish products necessitates the continuous production of current market data and trade standards. International environmental certification, applied to products approved under specific standards related to the traceability of fish and fish products, now requires the fishing industry to show that its practices are conducted sustainably (FAO, 2016b). In a national context in Canada, the fishing industry is increasingly involved with DFO in the co-production of fisheries scientific information, thereby facilitating the industry to meet the requirements of environmental certification bodies such as the Marine Stewardship Council.

5.2. Barriers to communication of information

Dissemination of information by the three organizations followed institutional norms rather than being dynamic in response to the changing information world. For instance, the organizations used their websites as their main communication tool. National governments still experience difficulties in overcoming the “digital divide” and in supporting widespread policy networks and communities, given time, money, and staff constraints. DFO staff members were generally constrained in their communication with the public and all external communication had to be vetted by the Communications Branch to meet communication protocols (circa June 2014). Furthermore, direct effort to communicate scientific advice outside of the DFO CSAS and NAFO processes was not considered necessary by either organization, since the advice was being produced by scientists for managers for immediate fisheries decision-making. While DFO and NAFO made no formal use of social media in decision-making contexts, FAO was increasingly utilizing social media tools, such as Twitter, to communicate about its mandate and engage discussions with like-minded organizations and the public during international meetings.

External communication of FAO technical project information is pursued on an ad hoc basis and dependent on whether the transfer of information to external audiences was an explicit component of a project cycle. Constraints in communicating information were mainly due to the lack of staff, apart from specialists, assigned to communicate information to member country focal points and other stakeholders. Respondents believe that FAO is moving “opportunistically towards an advocacy organization” (FAO 24). FAO’s apparent adoption of an advocacy role is attributed to its dedication of considerable resources to produce publications that contain simple messages particularly relevant to national policy-making in developing countries. Such information products include, video, flyers, infographics, brochures, and booklets. A “generic” product or “one size fits all” type of publication is expected to be used by diverse audiences, including policy-makers, managers, scientists, and the general public. FAO staff referred to the audience as a “vague type of user” (FAO 30).

Different interpretations of uncertainty of advice by DFO and NAFO scientists and managers exist. Uncertainty in the fisheries advice is generally accepted by managers as part of the evidence. DFO fisheries scientists and managers participating in NAFO commented on the low profile of scientific uncertainty in discussions of the scientific advice during the briefing of delegations. One manager stated: “The Fisheries Commission does not question the veracity or credibility of the scientific information, they just want further clarification . . . if I am fishing over there which is worth millions and jobs, I have to look at the scientific advice more closely” (NAFO 17). In contrast, ecosystem scientists often have difficulty communicating with fisheries scientists about the benefits of incorporating ecosystem considerations in stock assessments. Consequently, ecosystem information is not

systematically incorporated into the advice and its use is perceived as merely contextual, i.e., “it is used as background information when ecosystem data is needed to describe a process in a stock assessment” (NAFO 2).

Aspects of the organizational structure of DFO and NAFO pose barriers to information movement. DFO-MR scientists perceived the separation of responsibilities of their region and the DFO headquarters (DFO-NCR) as a disconnect between the two administrative regions. As a federal government department, high level policy is developed in DFO’s national region and implemented in the other regions. These organizational aspects contribute to the asynchronous character of the science-policy interface, where national policy often precedes the available science needed to implement the policy in the regions. This characteristic was previously highlighted in Curran et al. (2012) and Rice (2005) with regard to attempts to adopt the ecosystem approach to fisheries management in DFO. The structure of the DFO CSAS and the NAFO request for advice processes favour traditional single-species fisheries management and limits communication of ecosystem advice.

Aspects of the DFO-CSAS and the NAFO FC request for advice processes also create barriers in the information pathways. “Matching-up” of science and management may be dangerous as other sources of relevant information and issues may be overlooked, for example, climate change considerations related to fisheries. As a senior scientist stated:

We have given up the ability to say this is the scope of advice that you really need with regard to that question. We let the operational sectors and the policy sectors decide the breadth that they want science to consider and we are supposed to stop with their definition. We have lost something of the potential value of science if we constrain the range of questions to only those questions that the policy and management want to consider. (DFO 21)

Furthermore, given the composition of the DFO CSAS meetings, the scientific rigour of this peer review process is thought to be at a lower level than if the review was conducted by fisheries or ecosystem scientists only. As a senior scientist stated:

Expansion of the peer review to include academics, industry, and NGOs in the CSAS process is a watered down version of peer review. Multi-stakeholder participation in CSAS meetings does not allow for addition of any technical details and more in-depth discussions. (DFO 21)

5.3. Enablers to communication of information

Overlapping memberships of science and management groups in DFO, NAFO, and FAO illustrate the many science-policy interfaces that facilitate communication within and between the organizations (Fig. 3).

As a Contracting Party to NAFO, the interface between the Canadian organization (DFO) and NAFO is particularly important as most managers and national scientists described their work in NAFO as an extension of their national duties. Consequently, the success of NAFO as a regional fisheries management body is largely determined by its Contracting Parties. Likewise, FAO as an authority on fisheries statistics depends on the contribution of data by member countries. DFO’s scientists, fisheries managers, and policy-makers participating in NAFO and FAO meetings act as both the drivers to work completed as well as the implementers of the decisions in the national context. This overlapping membership is illustrated in structures dedicated to providing advice for management of a range of issues, such as deep-sea fisheries, where the intersecting institutional membership facilitated the flow of

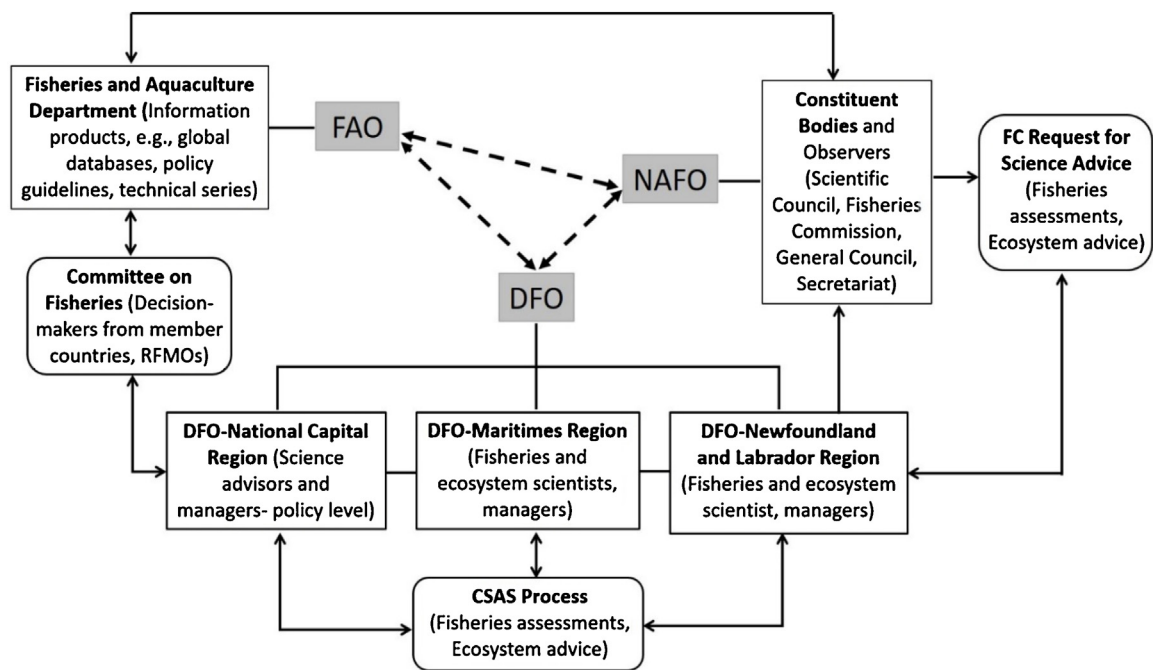


Fig. 3. The general interplay among DFO, NAFO, and FAO – a dynamic fisheries information network underpinning policy and decision-making.

related information produced by DFO, NAFO, and FAO, into a global database on VMEs (FAO, 2016d). The formation of new joint NAFO Fisheries Commission-Scientific Council working groups, and scientific working groups, such as the Working Group on Ecosystem Science and Assessments, resulted in important fora to facilitate communication between scientists and managers and between fisheries and ecosystem scientists. Overlapping memberships of scientific and policy working groups among the three organizations also demonstrate important information pathways that can be utilised to support implementation of ecosystem-based management.

The trade-offs in applying the attributes of information (credibility, relevance, and legitimacy) support communication of information, its use in policy- and decision-making, and the fulfilment of the respective fisheries management mandates (Table 3). The trade-offs likely developed over time in response to external factors and the organizations may not be conscious of their existence. Trade-offs among the attributes do not exist in isolation; they are interrelated, dynamic, and are contingent on the decision-making (linear and collaborative) and information use (linear and continuum) models, issues, and the actors. The origin of the “matching-up” trade-off associated with the DFO CSAS and the NAFO request for advice processes may be found in the history of decision-making in the organizations in the 1980s and 1990s when overfishing and collapsing fisheries were intensely disputed in the Northwest Atlantic (Lear and Parsons, 1993). The CSAS process developed from the Canadian Atlantic Fisheries Scientific Advisory Committee (CAFSAC) process – disbanded in 1992 after the collapse of the northern cod fishery – partly due to lessons learnt from the latter process. The “matching-up” trade off in NAFO guards against intense disagreements and acrimonious relationships that were common among NAFO Contracting Parties between 1979 and 1994 (Barry et al., 2014; NAFO staff, personal communication, October 2013). Agreements in NAFO are now developed by consensus, which represents a substantial shift in the management and policy-making philosophies of the NAFO Contracting Parties. The “consultation-peer review” trade-off seen in the DFO-CSAS

process likely developed in response to global efforts to implement ecosystem-based management which promote the involvement of diverse stakeholder groups in management. Legitimacy of the information was preferred over credibility when the robust science review, characteristic of the CAFSAC process, was replaced with the wide-peer review community in the CSAS process.

Communication between fisheries scientists and managers has subsequently increased, after the CAFSAC process was discontinued, and now managers often come with science backgrounds and a better understanding of technical fisheries management concepts. Similarly, long relationships between scientists and managers have engendered a level of trust that facilitates dialogue and scientists are more receptive to management needs. The enhanced trust may also explain why uncertainty related to fisheries scientific information, e.g., in stock assessments, was not considered to be a critical barrier to decision-making in DFO and NAFO. A manager described a serendipitous element to this improved communication related to personalities:

We are very fortunate here to have managers who have that philosophy because it is not a given ... there is nothing institutionalised about why this works. It works because you have people in levels who think this is the way that it should work ... it could just as easily not work. (DFO 10)

6. Conclusions and recommendations

This research encompassing interlinked fisheries management organizations that operate at different spatial scales identified five significant characteristics of information use in decision-making at the science-policy interface:

- A well-defined process for producing scientific advice and making management decisions embedded in two organizations – DFO and NAFO – is essential for producing credible, relevant, and legitimate information for decision-making.
- FAO functions as a boundary organization as it bridges scientific and policy-making communities in member countries.

- National, regional, and international policy- and decision-making rely heavily on publications produced by governmental organizations (grey literature).
- Multiple actors, apart from governmental scientists and managers, play increasingly important roles in policy- and decision-making processes.
- Trade-offs in the attention given to attributes of information, i.e., credibility, relevance, and legitimacy, are necessary and are always in action at the interface so that the fisheries organizations can be successful in meeting their objectives (e.g., management, governance)

This research empirically demonstrates the important and complex role of information at the science-policy interface. The results are especially relevant given the increasing global concern about an apparent disconnect between the information produced by scientists and its use by decision-makers (UN DESA, 2013). Furthermore, the results support the collective call for knowledge-making processes to be recognised as integral parts of governance regimes as described in Turnhout et al. (2016).

A general operational culture exists within governmental organizations that regards scientific information, contained in grey literature (e.g., technical reports) produced by their scientists, as the most credible and relevant compared to other sources. The preference for grey literature is largely due to its timeliness and applicability to decisions that need to be made. This recognition of the important role of grey literature warrants greater understanding and acceptance of its value in policy- and decision-making. The perception of an implicit or inconsequential role of information must be addressed, which was evident in one respondent's comments: "I don't think at all from the information point of view. I am thinking how to impact people's behavioural patterns and their motivations ... so I don't use the word information in my thinking at all" (FAO 20). Despite producing vast quantities of grey literature many governmental organizations generally also lack the capacity to conduct systematic evaluations of their function in fisheries management and policy- and decision-making. In order for resources to be used more effectively, evaluating the use (and influence) of information contained in the grey literature is necessary particularly in light of increasingly limited resources for its production and communication.

Empirical evidence from the case study of FAO (which describes itself as a knowledge organization) revealed characteristics of a boundary organization. Few studies of boundary organizations from an information management and a public policy perspective have been reported in the literature. FAO's organizational structure and information management processes (e.g., its principle of neutrality, staff who are specialists, and knowledge production through partnerships) promote communication between scientific and decision-making groups. The Intergovernmental Panel on Climate Change and the Intergovernmental Science-Policy Platform for Biodiversity and Ecosystem Services are prominent examples of global boundary organizations with similar characteristics to FAO (Löfmarck and Lidskog, 2017; Siebenhüner, 2003). NAFO, a regional fisheries management body does not operate as a formal boundary organization, although it plays a bridging or intermediary role between FAO and national scientists and managers in DFO by raising awareness of the international agendas in fisheries management. The characteristics of a boundary organization depend on the issues being addressed and the institutional landscape in which an organization operates.

This paper adopted a unique approach (mixed research methods applied within information production and decision-making processes) to explicitly examine the role of scientific information in governmental decision-making for fisheries management. This approach was particularly effective in revealing

challenges of implementing ecosystem management. The increasing pressure to adopt ecosystem approaches, which demands the inclusion of information from different disciplines (e.g., biological, economic, and social), complicates communication at the science-policy interface. Administrative structures in DFO and NAFO continue to support the traditional single species fisheries management approach and the reliance on science – excluding other types of knowledge – for policy-making (Curran et al., 2012; NAFO, 2011). Governmental organizations frequently face fiscal constraints that limit hiring additional staff, e.g., more ecosystem scientists and social scientists, to address the need for additional information. The inherent uncertainty related to fisheries advice is addressed more readily than uncertainty associated with ecosystem advice. Organizations have not yet fully modified decision-making processes to accommodate ecosystem-based management. Decision-making structures are difficult to fully change, but steps can be taken to achieve implementation goals based on what is learnt, e.g., DFO and NAFO demonstrate varying levels of organizational restructuring to promote discussions among scientists and managers and to integrate fisheries and ecosystem concerns.

The information pathways in operational decision-making encompass the presence of multiple, dynamic science-policy interfaces within DFO, NAFO, and FAO. Knowledge of the information pathways can guide each of the organizations in evaluating or modifying their production and communication practices to maintain the standards of credibility, relevance, and legitimacy of their information, and to balance the trade-offs among these attributes. Other governmental organizations, research scientists in academic settings, NGOs, and other actors who wish to contribute but are unfamiliar with the ways in which scientific information is communicated in DFO, NAFO, and FAO will benefit from the results reported here. Similar to these three organizations, many others rely on their own publications as a primary means for communicating the results of research and related activities. The methods used in this research and the resulting new knowledge will provide direction for future studies aimed at gaining greater understanding of information pathways in other fisheries and resource or environmental management fields.

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